Jaejin Cho, Ph.D.

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RESEARCH INTERESTS

- **Machine Learning:** Physics-guided Deep Learning, Self-Supervised Learning, Super-resolution
- **Medical Imaging:** Image Reconstruction, Quantitative Imaging, Pulse Sequence Design, Parallel Imaging, MRI Physics
- **Image Processing:** Numerical Optimization, Low-Rank Reconstruction, Compressed Sensing

EXPERIENCE

Sejong University — **Mar. 2024 – Current**
Assistant Professor, Department of Artificial Intelligence and Robotics — *Seoul, South Korea*

Massachusetts General Hospital (MGH) / Harvard Medical School — **Nov. 2018 – Jan. 2024**
Research Fellow, Athinoula A. Martinos Center for Biomedical Imaging — *Charlestown, MA, USA*

- Developed neural networks for rapid MRI acquisition and reconstruction
- Principal Investigator: Dr. Berkin Bilgic

EDUCATION

KAIST (Korea Advanced Institute of Science and Technology) — **Mar. 2014 – Feb. 2019**
Ph.D. in Electrical Engineering — *Daejeon, South Korea*

- Thesis: Water-Fat Separated Image Reconstruction for Bipolar Multi-Echo GRE Imaging in MRI
- Advisor: Dr. HyunWook Park

KAIST — **Feb. 2012 – Feb. 2014**
M.S. in Electrical Engineering — *Daejeon, South Korea*

- Thesis: Off-resonance Artifact Reduction Using the Echo-time Encoding Technique in MRI
- Advisor: Dr. HyunWook Park

Korea University — **Mar. 2008 – Feb. 2012**
B.S. in Electrical Engineering — *Seoul, South Korea*

RESEARCH GRANTS & PROJECTS

Principal Investigator

National Research Foundation of Korea (NRF) — **Apr. 2025 – Current**
Next-Generation Innovative Medical Imaging Technique Development

- **Total Budget:** $\approx$ \$492,500 (KRW 728M)
- **Grant No.:** RS-2025-00555277
- **Title:** Development of a Next-Generation Innovative Medical Imaging Technique Leveraging Motion Correction and Synthesis in High-Resolution, Accelerated Quantitative MRI

Participating Researcher

MIT-Korea – KAIST Seed Fund **Jan. 2019 – Aug. 2020**
Investigator (PIs: H.W. Park, E. Adalsteinsson) *Topic: Diffusion Magnetic Resonance Imaging*

MSIT & IITP (ICT R&D Program) **Jul. 2017 – Dec. 2018**
Graduate Researcher *Topic: Machine learning and statistical inference framework for explainable AI*

National Research Foundation of Korea (NRF) **Jul. 2014 – Feb. 2019**
Graduate Researcher (Brain Research Program) *Topic: Multi-channel/multi-nuclei MR imaging and reconstruction method*

MI Korea **Jul. 2013 – Jun. 2014**
Graduate Researcher *Topic: Development of Silicon based photomultiplier*

NRF & MSIT **Jun. 2013 – May 2014**
Graduate Researcher *Topic: Perfusion and Diffusion Weighted MRI Methods for Early Diagnosis*

Ministry of Science and ICT (MSIT) **May 2013 – Apr. 2014**
Graduate Researcher *Topic: Elucidation of multiple intentions through multi-modal imaging and analysis*

Ministry of Science and ICT (MSIT) **Mar. 2013 – Feb. 2014**
Graduate Researcher *Topic: Measurement and analysis of high-resolution multimodal brain functional information*

Ministry of Science and ICT (MSIT) **Sep. 2012 – Aug. 2013**
Graduate Researcher *Topic: Smart Neural Sensor Network System*

HONORS & AWARDS

- **Editor's Choice Articles**, Bioengineering Journal (Special Issue: AI in MRI)
- **ISMRM Merit Award – Summa cum laude** (Top 5% review score, 1st author), 2016
- **ISMRM Merit Award – Magna cum laude** (Top 15% review score, 1st author), 2015
- **ISMRM Student Stipend**, 2015–2017
- **Graduate Scholarship for Doctoral Course**, Government of the Republic of Korea, 2014–2018
- **Graduate Scholarship for Master Course**, Government of the Republic of Korea, 2012–2014
- **National Science and Engineering Scholarship**, Korea Student Aid Foundation, 2010–2011
- **Honors Scholarships**, Korea University, 2009 (Mar. & Sep.)

PEER-REVIEWED JOURNAL ARTICLES

** Full publication list available on Google Scholar.*

[1] **Jaejin Cho**, "Logarithmic Scaling of Loss Functions for Enhanced Self-Supervised Accelerated MRI Reconstruction". *Diagnostics* 15.23 (2025): 2993. https://doi.org/10.3390/diagnostics15232993

[2] Uten Yarach, Itthi Chatnuntawech, Congyu Liao, Surat Teerapittayanon, Siddharth Srinivasan Iyer, Tae Hyung Kim, Justin Haldar, **Jaejin Cho**, Berkin Bilgic, Yuxin Hu, Brian Hargreaves, Kawin Setsompop, "Blip-up blip-down circular EPI (BUDA-cEPI) for distortion-free dMRI with rapid unrolled deep learning reconstruction". *Magnetic Resonance Imaging* 115 (2025): 110277. https://doi.org/10.1016/j.mri.2024.110277

[3] So Hyun Kang, Jihoo Kim, **Jaejin Cho**, Clarissa Zimmerman Cooley, Amir Heydari, Berkin Bilgic, Tae Hyung Kim, "Enhanced Partial Fourier MRI With Zero-Shot Deep Untrained Priors". *IEEE Access* 12 (2024): 182157–182170. https://doi.org/10.1109/ACCESS.2024.3508761

[4] Yohan Jun, Yamin Arefeen, **Jaejin Cho**, Shohei Fujita, Xiaoqing Wang, P. Ellen Grant, Borjan Gagoski, Camilo Jaimes, Michael S. Gee, Berkin Bilgic, "Zero-DeepSub: Zero-shot deep subspace reconstruction for rapid

multiparametric quantitative MRI using 3D-QALAS". *Magnetic Resonance in Medicine* 91.6 (2024): 2459–2482. https://doi.org/10.1002/mrm.30018

[5] **Jaejin Cho**†, Borjan Gagoski†, Taehyung Kim, Fuyixue Wang, Daniel Nico Splitthoff, Wei-Ching Lo, Wei Liu, Daniel Polak, Stephen Cauley, Kawin Setsompop, P. Ellen Grant, Berkin Bilgic, "Time-efficient, High Resolution 3T Whole Brain Quantitative Relaxometry using 3D-QALAS with Wave-CAIPI Readouts". *Magnetic Resonance in Medicine* 91.2 (2024): 630–639. https://doi.org/10.1002/mrm.29865

[6] Yohan Jun, **Jaejin Cho**, Xiaoqing Wang, Michael Gee, P. Ellen Grant, Berkin Bilgic, Borjan Gagoski, "SSL-QALAS: Self-Supervised Learning for Rapid Multiparameter Estimation in Quantitative MRI Using 3D-QALAS". *Magnetic Resonance in Medicine* 90.5 (2023): 2019–2032. https://doi.org/10.1002/mrm.29786

[7] Zhifeng Chen, Congyu Liao, Xiaozhi Cao, Benedikt A Poser, Zhongbiao Xu, Wei-Ching Lo, Manyi Wen, **Jaejin Cho**, Qiyuan Tian, Yaohui Wang, Yanqiu Feng, Ling Xia, Wufan Chen, Feng Liu, Berkin Bilgic, "3D-EPI Blip-Up/Down Acquisition (BUDA) with CAIPI and Joint Hankel Structured Low-Rank Reconstruction for Rapid Distortion-Free High-Resolution T2* Mapping". *Magnetic Resonance in Medicine* 89.5 (2023): 1961–1974. https://doi.org/10.1002/mrm.29578

[8] **Jaejin Cho**, Borjan Gagoski, Taehyung Kim, Qiyuan Tian, Stephen Robert Frost, Itthi Chatnuntawech, Berkin Bilgic, "Wave-Encoded Model-based Deep Learning for Highly Accelerated Imaging with Joint Reconstruction". *Bioengineering* 9.12 (2022): 736. https://doi.org/10.3390/bioengineering9120736
(Selected as Editor's Choice Articles, Special Issue - AI in MRI)

[9] **Jaejin Cho**, Congyu Liao, Qiyuan Tian, Zijing Zhang, Jinmin Xu, Wei-Ching Lo, Benedikt A. Poser et al., "Highly Accelerated EPI with Wave Encoding and Multi-shot Simultaneous Multi-Slice Imaging". *Magnetic Resonance in Medicine* 88.3 (2022): 1180–1197. https://doi.org/10.1002/mrm.29291

[10] Zijing Zhang, Long Wang, **Jaejin Cho**, Congyu Liao, Hyeong-Geol Shin, Xiaozhi Cao, Jongho Lee et al., "BUDA-SAGE with self-supervised denoising enables fast, distortion-free, high-resolution T2, T2*, para- and dia-magnetic susceptibility mapping". *Magnetic Resonance in Medicine* 88.2 (2022): 633–650. https://doi.org/10.1002/mrm.29219

[11] Yamin Arefeen, Onur Beker, **Jaejin Cho**, Heng Yu, Elfar Adalsteinsson, Berkin Bilgic, "Scan-specific artifact reduction in k-space (SPARK) neural networks synergize with physics-based reconstruction to accelerate MRI". *Magnetic Resonance in Medicine* 87.2 (2022): 764–780. https://doi.org/10.1002/mrm.29036

[12] **Jaejin Cho**, HyunWook Park, "Robust water–fat separation for multi-echo gradient-recalled echo sequence using convolutional neural network". *Magnetic Resonance in Medicine* 82.1 (2019): 476–484. https://doi.org/10.1002/mrm.27697

[13] **Jaejin Cho**, HyunWook Park, "Interleaved bipolar acquisition and low-rank reconstruction for water–fat separation in MRI". *Medical Physics* 45.7 (2018): 3229–3237. https://doi.org/10.1002/mp.12981

[14] Dongchan Kim, Hyunseok Seo, **Jaejin Cho**, Kinam Kwon, HyunWook Park, "Non-contrast-enhanced peripheral MR angiography using velocity-selective excitation". *Magnetic Resonance in Medicine* 79.2 (2018): 779–788. https://doi.org/10.1002/mrm.26732

REFEREED CONFERENCE PROCEEDINGS (FULL PAPERS)

* *For abstracts and presentations, see "Invited Talks & Selected Abstracts" section.*

[1] **Jaejin Cho**, Yohan Jun, Xiaoqing Wang, Caique Kobayashi, Berkin Bilgic. "Improved Multi-Shot Diffusion-Weighted MRI with Zero-Shot Self-Supervised Learning Reconstruction". *Medical Image Computing and Computer Assisted Intervention – MICCAI 2023*. Lecture Notes in Computer Science, vol 14220. Springer, Cham. https://doi.org/10.1007/978-3-031-43907-0_44

[2] Congyu Liao, Xiaozhi Cao, **Jaejin Cho**, Zijing Zhang, Kawin Setsompop, Berkin Bilgic. "Highly efficient MRI through multi-shot echo planar imaging". *Wavelets and Sparsity XVIII*. Vol. 11138. International Society for

Optics and Photonics, 2019. https://doi.org/10.1117/12.2527183

PATENTS

[1] Berkin Bilgic, Zijing Zhang, Congyu Liao, **Jaejin Cho**, Mary Kate Manhard. "Distortion-free diffusion and quantitative magnetic resonance imaging with blip up-down acquisition of spin-and gradient-echoes". *U.S. Patent Publication No. US12487307B2*.

[2] HyunWook Park, **Jaejin Cho**. "A magnetic resonance imaging apparatus and a method for controlling the same". *U.S. Patent Application No. US20180210059; KR Patent Application No. KR20180085976A*.

[3] Young-Beom Kim, HyunWook Park, Dongchan Kim, Hyunseok Seo, Kinam Kwon, **Jaejin Cho**. "Magnetic resonance imaging apparatus and method". *U.S. Patent Application No. US20170131377A1; EP Patent Application No. EP3168636A3; KR Patent Application No. 10-2015-0157487*.

INVITED TALKS & SELECTED ABSTRACTS

Selected Invited Lectures

- **Jaejin Cho**. "Diffusion Artifact and Correction". *ISMRM 31st Annual Meeting*, London, UK, 2022.

Selected Oral Presentations (Conference Abstracts)

[1] **Jaejin Cho**, Borjan Gagoski, Tae Hyung Kim, Qiyuan Tian, Stephen Robert Frost, Itthi Chatnuntawech, Berkin Bilgic. "Rapid Quantitative Imaging Using Wave-Encoded Model-Based Deep Learning for Joint Reconstruction". *ISMRM 31st Annual Meeting*, London, 2022, pp. 4315.

[2] **Jaejin Cho**, Tae Hyung Kim, Borjan Gagoski, Zijing Zhang, Avery JL Berman, Congyu Liao, Berkin Bilgic. "Variable Flip, Blip-Up and -Down Undersampling (VUDU) Enables Motion-Robust, Distortion-Free Multi-Shot EPI". *ISMRM 31st Annual Meeting*, London, 2022, pp. 6000.

[3] **Jaejin Cho**, Avery Berman, Borjan Gagoski, Congyu Liao, Jason Stockmann, Jonathan Polimeni, Berkin Bilgic. "VUDU: motion-robust, distortion-free multi-shot EPI". *ISMRM 29th Annual Meeting*, Online, 2021, pp. 0002.

[4] **Jaejin Cho**, Dongchan Kim, Hyunseok Seo, Kinam Kwon, Seohee So, HyunWook Park. "Correction of Chemical-Shift Ghost Artifact in Blipped Controlled Aliasing Parallel Imaging". *ISMRM 24th Annual Meeting*, Singapore, 2016, pp. 0610. *(Recipient of a Summa Cum Laude ISMRM Merit Award)*

[5] **Jaejin Cho**, HyunWook Park. "A Fast Multi-Echo Imaging for Water-Fat Separation Using Low Rank Property". *The 4th International Congress on Magnetic Resonance Imaging (ICMRI)*, 2016, pp. SS04-9.

ACADEMIC SERVICE

- **Moderator:** ISMRM-endorsed Workshop (2021) – MRI Acquisition and Reconstruction
- **Journal Reviewer:** Magnetic Resonance in Medicine, Medical Physics, IEEE Transactions on Computational Imaging, IEEE Journal of Selected Topics in Signal Processing
- **Membership:** ISMRM (Member), KSMRM (Board Member)

TECHNICAL SKILLS

- **Programming:** C, C++, MATLAB, Python, TensorFlow, Keras
- **Medical Tools:** IDEA (Siemens sequence coding), FSL, Freesurfer, MRtrix
- **Open Source:** https://github.com/jaejin-cho